

Jiacheng Yao

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EDUCATION

- **Beijing Normal-Hong Kong Baptist University (BNBU)** Zhuhai, China
BSc (Hons) in Applied Mathematics (English-medium) *Sep. 2023 – Jun. 2027*
- **Hong Kong Baptist University (HKBU)** Hong Kong SAR, China
Exchange Student *Jan. 2026 – May 2026*
- **Relevant Coursework:** Data Structures and Algorithms, Probability Theory, Optimization, Numerical Analysis, Multivariate Analysis, Big Data Analytics, Data Mining and Knowledge Discovery, Artificial Intelligence Workshop

HONORS & AWARDS

- **Second-Class Student Scholarship** — AY 2023–2024, 2024–2025
- **Third Prize, Guangdong Contemporary Undergraduate Mathematical Contest in Modeling 2025**
- **Scholarship for the Extended Study Programme in the Greater Bay Area** — HKBU, Jan. 2026

RESEARCH EXPERIENCE

- **Neural Curvature Estimation for Two-Phase Flow Simulations**
Supervisor: Dr. Jiaqi Zhang (BNBU; Beijing Normal University, Zhuhai Campus) *Jan. 2026 – Present*
 - **Method development:** Built the data and training pipeline for neural curvature estimation, generating 83.5 million labeled samples at five grid resolutions from analytically defined circular and elliptical interfaces and training a separate neural network for each resolution using local level-set and normal-vector stencils.
 - **Quantitative evaluation:** Benchmarked the neural models against second-order finite-difference curvature estimation; when training and test resolutions were matched, the models reduced MSE by 5.6–260 $\times$ across the five test resolutions. Evaluated cross-resolution transfer, sensitivity to level-set reinitialization, and generalization to unseen flower-shaped interfaces.
 - **Solver deployment:** Converted the trained networks into standalone C routines and integrated them directly into Basilisk's CLSVOF solver without external machine-learning dependencies. Tested the modified solver on stationary bubble, capillary wave, and rising bubble benchmarks.
 - **Research output:** First-author manuscript in preparation on neural curvature estimation for level-set two-phase flow simulations.
- **Automated Arrow-Impact Localization and Ring-Score Recognition**
Supervisors: Dr. Lidong Fang (SUFU); Prof. Huaxiong Huang (DKU) *Jun. 2026 – Present*
 - Developed and implemented an end-to-end computer vision pipeline extending earlier intelligent arrow-selection work toward an automated archery training and assessment system, including data curation and annotation, training strategy design, model training, and evaluation.
 - Explored an initial ResNet-based formulation before adopting YOLO26x as the current baseline for arrow-impact localization. Trained the baseline on annotated public data and evaluated it on self-collected target images to assess generalization from public datasets to practical shooting conditions.
 - Evaluated global image preprocessing and augmentation, including contrast adjustment and sharpening, to improve robustness to real-world imaging variation; the best preliminary configuration achieved both precision and recall above 70% on self-collected target images.

PROFESSIONAL EXPERIENCE

- **Research and Development of Third-Generation Intelligent Arrow Selection Equipment and Systems**

Research Assistant / Faculty of Science and Technology, BNB

Jul. 2025 – May 2026

- Developed and carried out a standardized experimental workflow for a joint arrow-selection project with the 18th Institute of CASC and Beijing Institute of Technology for the Chinese National Archery Team, including controlled shooting tests, multi-orientation shaft-deflection measurements, and systematic recording of arrow and impact data.
- Developed a high-speed-video analysis workflow to track arrow motion at release and extract launch velocity, launch angle, and trajectory-deviation measures for individual shots.
- Built and manually validated a structured dataset linking physical arrow measurements, shot-level kinematics, ring scores, and impact locations; analyzed impact dispersion, scoring consistency, and cross-round variability to support quantitative comparison and selection of arrows for competition.
- **Co-authored** Arrow Selection for Elite Series Arrows: Data-Driven Archery Arrow Selection and **contributed equally to the work**; manuscript submitted to Sports Engineering.

- **Wanhoo Technology (Shenzhen) Co., Ltd.**

Remote Research Intern / Supervisor: Dr. Jiaqi Zhang

Sep. 2025 – Mar. 2026

- Extended a RAG-based platform to build a teacher-facing AI education system for lesson preparation, assessment generation, and the creation of editable Word and PowerPoint teaching materials.
- Enhanced the document-ingestion and retrieval layer using MinerU/OCR to structure teaching materials as Markdown and JSON, and integrated embedding, reranking, and chat models through SiliconFlow and Ollama for grounded retrieval and generation.
- Implemented an editable assessment workflow supporting five question types across three difficulty levels, natural-language revision, question-level editing, and DOCX export; integrated MCP-based tool execution for multi-step tasks and deployed the system internally with Docker.

LEADERSHIP & CAMPUS ACTIVITIES

- **Youth League Branch, Faculty of Science and Technology, BNB**

Publicity Committee Member

Sep. 2023 – Present

- Helped organize and publicize branch activities and maintained regular communication with members about their academic progress and general well-being.

- **Dream Safari Charity Club, BNB**

Core Member & Volunteer

Sep. 2023 – Jun. 2024

- Coordinated publicity for club activities and volunteered in rural outreach activities involving photography and educational support; helped document program activities and was recognized as an Excellent Volunteer.

SKILLS

- **Technical:** Python, C, MATLAB, SQL, Linux, Docker, L^AT_EX.
- **Languages:** Chinese (native); English.